

2.5.5

26.07.02 (Thu).

①

Negative Binomial Distribution.

$$f(x; r, p) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \quad (x=r, r+1, \dots)$$

"Success prob: p , repeat up to r times success. (indep.)"

Denote $X \sim \text{NB}(r, p)$.

$$E[X] = r E[X_1] = \frac{r}{p}$$

$$\text{Var}(X) = r \text{Var}(X_1)$$

$$= \frac{r q}{p^2} \quad (p+q=1).$$

$$\left(\begin{array}{l} \text{Let } X_1, X_2, \dots, X_r \sim \text{Geo}(p), \\ X = X_1 + X_2 + \dots + X_r. \end{array} \right)$$

Note. For constant r s.t. $r \leq n$, $X \sim \text{NB}(r, p)$ and $Y \sim \text{NB}(n, p)$.

If $X \leq n$, then

$$P(X \leq n) = P(Y \geq r) = \sum_{k=r}^n \binom{n}{k} p^k q^{n-k}$$

2.5.6 Hypergeometric distribution.



$\rightarrow$ r Red balls
 w White balls $\Rightarrow n$ balls, sampling without replacement.

X : # (red balls). $\rightarrow$ pdf of X is:

$$f(x) = \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}}, \quad x = \max(0, n-N+r), \dots, \min(r, n).$$

Denote $X \sim \text{HYP}(n, N, r)$.

②

$$\begin{aligned}
 E[X] &= \sum_{x=0}^n x \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}} = n \frac{r}{N} \sum_{x=1}^n \frac{\binom{r-1}{x-1} \binom{N-r}{n-x}}{\binom{N-1}{n-1}} \\
 &= n \frac{r}{N} \sum_{y=0}^{n-1} \frac{\binom{r-1}{y} \binom{N-1-r+1}{n-1-y}}{\binom{N-1}{n-1}} = n \frac{r}{N} \cdot 1.
 \end{aligned}$$

$$\begin{aligned}
 E[X(X-1)] &= \sum_{x=0}^n x(x-1) \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}} \\
 &= n(n-1) \frac{r(r-1)}{N(N-1)} \sum_{x=2}^n \frac{\binom{r-2}{x-2} \binom{N-r}{n-x}}{\binom{N-2}{n-2}} \\
 &= n(n-1) \frac{r(r-1)}{N(N-1)} \sum_{y=0}^{n-2} \frac{\binom{r-2}{y} \binom{N-2-r+2}{n-2-y}}{\binom{N-2}{n-2}} \\
 &= n(n-1) \frac{r(r-1)}{N(N-1)} \cdot 1.
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \text{Var}(X) &= E[X^2] - (E[X])^2 = E[X(X-1)] + E[X] - (E[X])^2 \\
 &= n(n-1) \frac{r(r-1)}{N(N-1)} + n \frac{r}{N} - n^2 \frac{r^2}{N^2} \\
 &= n \frac{r}{N} \left[(n-1) \frac{r-1}{N-1} + 1 - \frac{nr}{N} \right] = \frac{nr}{N} \left[\frac{(N-r)(N-n)}{N(N-1)} \right].
 \end{aligned}$$

If $X \sim \text{HYP}(n, N, r)$ and $N \rightarrow \infty, r \rightarrow \infty, \frac{r}{N} \rightarrow p$, then

$$\lim_{\substack{N \rightarrow \infty \\ r \rightarrow \infty \\ \frac{r}{N} \rightarrow p}} \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}} = \binom{n}{x} p^x (1-p)^{n-x} \quad \text{holds.}$$

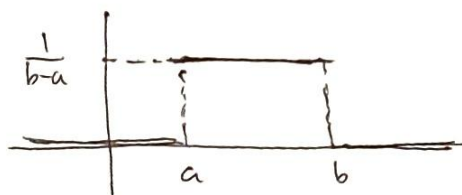
$\Rightarrow N \uparrow, r \uparrow \Rightarrow \text{Bin} \approx \text{HYP}$

2.6 Prob. Dist. of Conti. R.V.Def. 2.12

If a r.v. X satisfies $P(X=x)=0$ for any $x \in \mathbb{R}$, then X is a "Continuous random variable".

2.6.1 Uniform dist.

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{o.w.} \end{cases}$$



$$X \sim U(a, b).$$

$$F(x) = \int_{-\infty}^x f(y) dy = \int_{-\infty}^x \frac{1}{b-a} I_{(a,b)}(y) dy$$

$$= \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$$

$$E[X] = \int_a^b \frac{x}{b-a} dx = \frac{b+a}{2}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \int_a^b \frac{x^2}{b-a} dx - \left(\frac{b+a}{2}\right)^2 = \frac{(b-a)^2}{12}$$

2.6.2 Exponential dist.

Interarrival time W , for $w > 0$,

$$\begin{aligned} F(w) &= P(W \leq w) \\ &= 1 - P[\text{interval } [0, w) \rightarrow \overset{\text{event}}{A} \text{ does not occur.}] \\ &= 1 - e^{-\lambda w} \end{aligned}$$

$$\textcircled{f}) \quad f_w(w) = \frac{d}{dw} F(w) = \lambda e^{-\lambda w}, \quad w > 0.$$

$$\text{Let } \theta = \frac{1}{\lambda}. \Rightarrow f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad (x > 0). \quad \left(\text{Denote, } X \sim \text{EXP}(\theta) \right)$$

$$F(x) = \int_0^x \frac{e^{-\frac{t}{\theta}}}{\theta} dt.$$

$$= 1 - e^{-\frac{x}{\theta}}.$$

$$E[X] = \int_0^{\infty} x \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$$

$$= -x e^{-\frac{x}{\theta}} \Big|_0^{\infty} + \int_0^{\infty} e^{-\frac{x}{\theta}} dx = -\theta e^{-\frac{x}{\theta}} \Big|_0^{\infty} = \theta.$$

$$\text{Var}(X) = \int_0^{\infty} x^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx - \theta^2.$$

$$= -x^2 e^{-\frac{x}{\theta}} \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-\frac{x}{\theta}} dx - \theta^2$$

$$= 2\theta \int_0^{\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx - \theta^2 = 2\theta E[X] - \theta^2 = 2\theta^2 - \theta^2 = \theta^2.$$

Thm 2.19 (Memoryless Property)

$$X \sim \text{EXP}(\theta), \quad a > 0, t > 0.$$

$$P(X > a+t | X > a) = P(X > t).$$

$$\textcircled{pf}) \quad P(X > a+t | X > a) = \frac{P(X > a+t, X > a)}{P(X > a)} = \frac{P(X > a+t)}{P(X > a)}$$

$$= \frac{e^{-(a+t)/\theta}}{e^{-a/\theta}} = P(X > t). \quad \square$$